

Design of a Modular Digital Data Monitoring Unit with APB4 Bus Integration

From Standalone Hardware to SoC-Ready IP

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Abstract

Recent advances in embedded system technology have made real-time monitoring of sensor data increasingly important. While software polling is common, it often misses transient event spikes. Phase 1 of this project developed an 8-Bit Digital Event Monitoring Unit (DEMU) with user-defined thresholds, a "Sticky Alarm" mechanism, and a Fault Capture Register. However, for this IP to be usable in modern System-on-Chip (SoC) architectures, it requires a standardized processor interface. Therefore, in Phase 2, the DEMU core was integrated with an AMBA APB4 slave interface, transforming it into a memory-mapped, SoC-Ready IP. Additionally, a custom Python-based EDA automation tool (*AutoArchitect*) was developed for rapid RTL/Gate-level visualization.

Index Terms—Digital Event Monitoring, Verilog HDL, Sticky Alarm, APB4 Bus, SoC Integration, Python Automation.

1 Introduction

The rapid development of IoT and Industry 4.0 depends on intelligent hardware sensors. While processors can monitor data, they may miss critical transients due to software latency. Phase 1 addressed this by creating a dedicated hardware FSM watchdog. However, standalone hardware IPs require dynamic configuration by a CPU. Phase 2 bridges this gap by wrapping the DEMU core inside an APB4 protocol wrapper, allowing a processor to configure thresholds, read fault data, and acknowledge alarms via standard memory-mapped registers.

2 Problem Statement

2.1 Phase 1 Challenges (Group Project)

- 1) **Transitory Event Loss:** Software might miss brief voltage spikes.
- 2) **Insufficient Diagnostics:** Traditional alarms lack captured context (the fault value).
- 3) **Static Data Modes:** Inability to switch between signed/unsigned comparisons dynamically.

2.2 Phase 2 Challenges (Individual Project by Nigam Mehta)

- 1) **Hardwired Configuration:** The Phase 1 design required physical pins to change thresholds, which is impossible in a packaged SoC.
- 2) **No Processor Interfacing:** A CPU cannot read the fault capture register without a standardized bus protocol.
- 3) **Design Visualization:** Manually running Yosys commands to check hardware synthesis is tedious and error-prone.

3 System Architecture

The system follows a hierarchical RTL architecture, evolving from a flat design in Phase 1 to a wrapped SoC-IP in Phase 2.

3.1 Phase 1: DEMU Core Architecture

The core consists of seven functional sub-modules:

- **Sensor/Threshold Registers:** Synchronization and storage of 8-bit data.
- **Comparator:** Bitwise comparison logic (avoiding direct `</>` operators for strict hardware control).
- **FSM Controller:** 3-state Moore Machine (IDLE $\rightarrow$ ALARM $\rightarrow$ COOLDOWN).
- **Alarm & Fault Capture Registers:** Sticky latch and snapshot mechanisms.

3.2 Phase 2: APB4 Wrapper Architecture

To make the DEMU SoC-ready, an APB4 slave wrapper was designed around the core. The wrapper contains:

- **APB Interface Logic:** Handles `psel`, `penable`, `pwrite` handshaking.
- **Address Decoder:** Maps CPU addresses to DEMU internal signals.
- **Register Map:** Exposes DEMU configuration as memory-mapped registers (Table I).
- **Auto-Clear Logic:** Ensures the software acknowledgement pulse (`sw_ack`) lasts only one clock cycle.

Table 1: APB4 Memory-Mapped Register Map

Offset	Register	R/W	Description
0x00	THRESHOLD	R/W	Sets 8-bit threshold value
0x04	MODE_CTRL	R/W	Bit[0]: enable, Bit[1]: data_mode
0x08	SW_ACK	W	Write 1 to clear sticky alarm
0x0C	SENSOR_IN	R	Read live sensor data
0x10	ALARM_STAT	R	Bit[0]: alarm, Bits[8:1]: fault capture

4 Specifications

Table 2: System Specifications (Phase 1 & 2)

Parameter	Value / Description
Clock Frequency	50 MHz (Average FPGA Max)
Interface Type	AMBA APB4 (Phase 2) / Direct Pins (Phase 1)
FSM States	IDLE, ALARM, COOLDOWN
Data Width	8-bit Sensor / 8-bit Threshold
Address Width	32-bit APB Address Bus

5 Applications and SDG Mapping

The versatility of the DEMU IP allows deployment in diverse sectors mapped to UN SDGs: Industrial Motor Protection (SDG 9), Automotive Battery Management (SDG 7), Medical Fever Detection (SDG 3), and Environmental Flood Warning (SDG 11).

6 Implementation and Results

6.1 Phase 1: DEMU Core Verification

Verified via eight industrial application testbenches. The waveform analysis confirmed that the "ALARM OUT" holds HIGH and "FAULT CAPTURE" retains the trigger value even after "SENSOR DATA" returns to normal, proving the sticky alarm functionality.

6.2 Phase 2: APB4 Integration Verification

An APB-compatible testbench was developed using `apb_write` and `apb_read` tasks to simulate a CPU.

- **Threshold Configuration:** CPU successfully wrote 0x50 (80) to address 0x00.
- **Alarm Detection:** When sensor spiked to 90, CPU read address 0x10 and received 0x00B5 (Bit 0: Alarm=1, Bits[8:1]: Fault Value=90), confirming successful memory-mapped status readback.

6.3 AutoArchitect: EDA Automation Tool

To streamline the synthesis process, a Python script (`auto_architect.py`) was developed. It uses the `subprocess` module to automate Yosys, generates a JSON netlist, and calls `NetlistSVG` to render clean Gate-Level SVG diagrams instantly, eliminating manual command-line execution.

7 Individual Contribution Statement

- **Phase 1 (Group Project):** Design and simulation of the standalone DEMU core was a collaborative effort by Dhruv Bhanushalli, Nigam Mehta, and Vrushal Vaity.
- **Phase 2 (Individual Project):** The APB4 Wrapper architecture, Memory Map design, APB Testbench creation, and the *AutoArchitect* Python automation tool were entirely conceptualized, coded, and verified by **Nigam Mehta**.

8 Conclusion

Phase 1 successfully created a standalone Digital Event Monitor with zero-latency software notification and sticky alarm capabilities. Phase 2 successfully elevated this design into a SoC-Ready IP by integrating an AMBA APB4 interface, allowing dynamic processor control. The development of the AutoArchitect tool further demonstrated advanced

workflow automation skills. Future work includes integrating a Moving Average Filter for noisy environments and upgrading the bus interface to AXI4-Lite for higher bandwidth.

References

- [1] S. Brown and Z. Vranesic, “Fundamentals of Digital Logic with Verilog Design,” 2nd ed. McGraw-Hill, 2014.
- [2] M. D. Ciletti, “Advanced Digital Design with the Verilog HDL,” 2nd ed. Prentice Hall, 2010.
- [3] ARM Limited, “AMBA 3 APB Protocol Specification,” 2006.
- [4] C. Wolf, “Yosys Open SYnthesis Suite,” <http://www.clifford.at/yosys/>
- [5] United Nations, “Transforming our world: the 2030 Agenda for Sustainable Development,” 2015.